

Impact of Late First-Half Goals on Match Outcomes

2025-12-01

1. Data Loading and Initial Inspection We begin by loading the raw dataset events.csv. This step includes error handling to ensure the file exists and inspecting the raw structure of the data (column types, missing values) to determine necessary cleaning steps.

```
# Load data; file name is 'events.csv'
tryCatch({
  df <- readr::read_csv("events.csv")
  print("Data loaded successfully.")
}, error = function(e) {
  print("Error: Could not load the file.")
  df <- NULL
})
```

```
## [1] "Data loaded successfully."
```

```
# Missing Value Check
na_counts <- colSums(is.na(df))
na_counts_df <- data.frame(
  Variable = names(na_counts),
  Missing_Count = na_counts,
  Missing_Percentage = round((na_counts / nrow(df)) * 100, 2)
)
print(head(na_counts_df[na_counts_df$Missing_Count > 0, ]))
```

```
##                               Variable Missing_Count
## PartScores                    PartScores           7226
## ChampionshipSubGroupName      ChampionshipSubGroupName      904
##                               Missing_Percentage
## PartScores                    0.60
## ChampionshipSubGroupName      0.08
```

2. Data Cleaning and Type Conversion The raw data imports columns as characters or generic types. Here, we convert scores and periods to numeric types and parse the EventName to separate the competing teams. We also ensure the EventDate is properly formatted for any time-series analysis.

```
df_clean <- df %>%
  mutate(
    Score1 = as.numeric(Score1),
    Score2 = as.numeric(Score2),
    Period1 = as.numeric(Period1), # Goals scored in the 1st half
    Period2 = as.numeric(Period2), # Goals scored in the 2nd half
    FirstGoalTeam = as.numeric(FirstGoalTeam),
    FirstGoalMinute = as.numeric(FirstGoalMinute)
```

```

) %>%
separate(
  col = EventName,
  into = c("HomeTeam", "AwayTeam"),
  sep = " - ",
  remove = FALSE,
  extra = "merge"
) %>%
mutate(
  EventDate = lubridate::ymd_hms(EventDate)
)

cat("Cleaned dataframe dimensions:", dim(df_clean))

```

```
## Cleaned dataframe dimensions: 1197712 16
```

3. Feature Engineering: Identifying Late Goals We define a “Late First-Half Goal” as a goal scored between the 41st and 50th minute (inclusive of stoppage time).

```

# Function to check for a late first-half goal (41st min up to 50th min)
check_late_goal <- function(goal_minutes_string) {
  if (is.na(goal_minutes_string) || goal_minutes_string == "NULL") {
    return(0)
  }

  # Split multiple goals by comma
  raw_tokens <- unlist(strsplit(goal_minutes_string, ","))

  # Helper to parse "45+2" into 47
  parse_minute <- function(x) {
    if (grepl("\\+", x)) {
      parts <- as.numeric(unlist(strsplit(x, "\\+")))
      return(sum(parts))
    } else {
      return(as.numeric(x))
    }
  }

  minutes <- suppressWarnings(sapply(raw_tokens, parse_minute))

  # Check range
  if (any(minutes >= 41 & minutes <= 50, na.rm = TRUE)) {
    return(1)
  } else {
    return(0)
  }
}

df_features <- df_clean %>%
  rowwise() %>%
  mutate(
    LateFH_Goal_Scored = check_late_goal(Team1GoalMinutes),
    LateFH_Goal_Conceded = check_late_goal(Team2GoalMinutes)
  )

```

```

) %>%
ungroup() %>%
mutate(
  Match_Outcome = case_when(
    Score1 > Score2 ~ "Win",
    Score1 < Score2 ~ "Loss",
    Score1 == Score2 ~ "Draw",
    TRUE ~ NA_character_
  ),
  Match_Outcome = factor(Match_Outcome, levels = c("Draw", "Win", "Loss"))
)

```

4. Filtering for Top 5 Leagues To ensure consistency in competition level, we filter the dataset to include only matches from the "Top 5" European leagues. We handle multi-language variations (English and Russian/Cyrillic) commonly found in data scrapers.

```

# Filtering data
top_5_leagues_variants <- c(
  "Premier League", " - ",
  "Primera Division", " ", " ",
  " ", "Serie A",
  "Bundesliga", " ",
  "Ligue 1", " 1"
)

df_filtered <- df_features %>%
  filter(ChampionshipSubGroupName %in% top_5_leagues_variants) %>%
  mutate(
    Standard_League_Name = case_when(
      ChampionshipSubGroupName %in% c("Premier League", " - ") ~ "Premier League",
      ChampionshipSubGroupName %in% c("Primera Division", " ", " ") ~ "La Liga",
      ChampionshipSubGroupName %in% c(" ", "Serie A") ~ "Serie A",
      ChampionshipSubGroupName %in% c("Bundesliga", " ") ~ "Bundesliga",
      ChampionshipSubGroupName %in% c("Ligue 1", " 1") ~ "Ligue 1",
      TRUE ~ "Other"
    ),
    Standard_League_Name = factor(Standard_League_Name)
  )

cat("Matches remaining after filtering:", nrow(df_filtered))

```

```
## Matches remaining after filtering: 103587
```

5. Visual Analysis Before modeling, we visualize the data to understand the distribution of outcomes and the prevalence of late goals.

Distribution of Match Outcomes by League

```

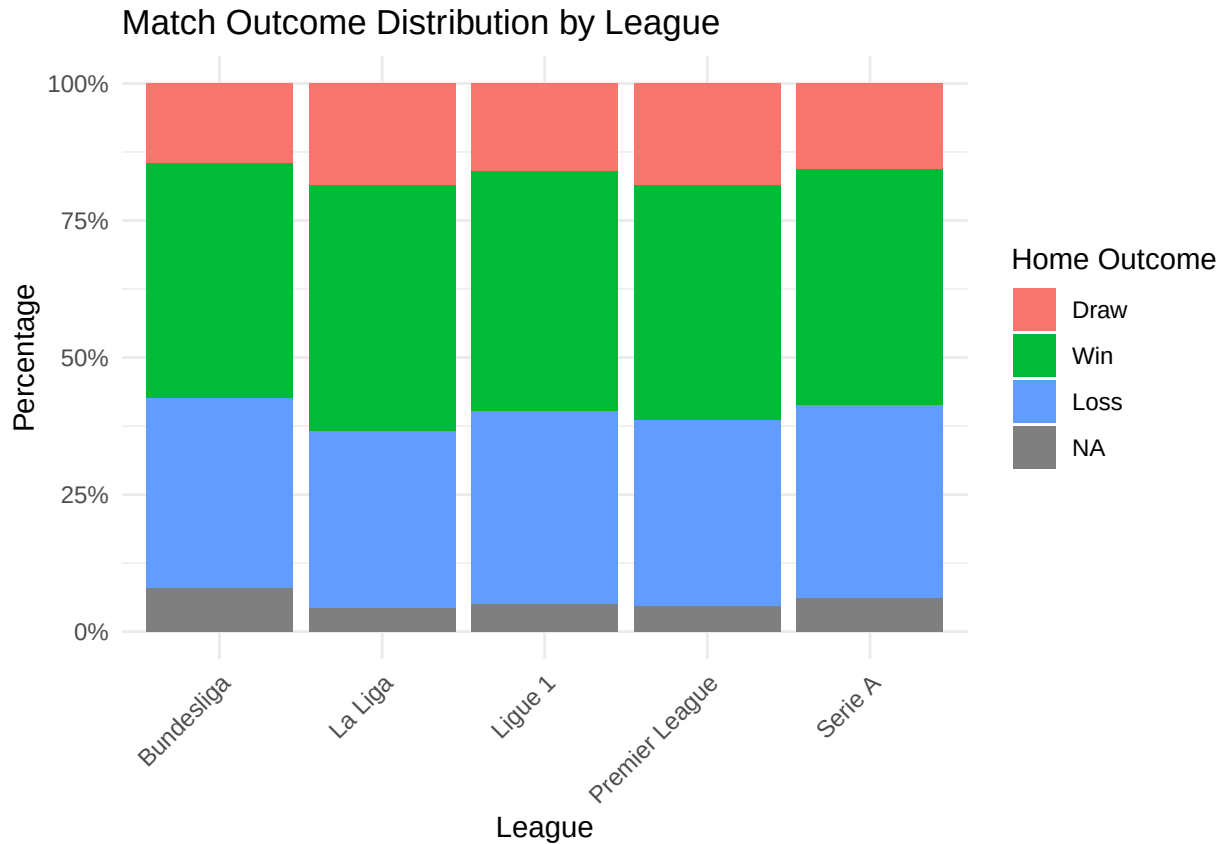
ggplot(df_filtered, aes(x = Standard_League_Name, fill = Match_Outcome)) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  labs(

```

```

title = "Match Outcome Distribution by League",
y = "Percentage",
x = "League",
fill = "Home Outcome"
) +
theme_minimal() +
theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



6. Statistical Modeling (Multinomial Regression) We use Multinomial Logistic Regression to model the probability of a Win, Loss, or Draw.

Model Note: We use the Home Team perspective for modeling. We do not duplicate the data (flipping to add Away rows) because this would violate the statistical assumption of independence (a Home Win guarantees an Away Loss in the same match).

Variables:

Late Goal Scored: Did the home team score in the 41-50 min window?

Late Goal Conceded: Did the away team score in that window?

Halftime Goals: Control variable for the current state of the match.

League: Control variable for league-specific trends.

```

# We use the Home Team perspective (df_filtered) directly
# Rename cols for clarity in the model output
df_model <- df_filtered %>%
  mutate(
    Home_Late_Goal = LateFH_Goal_Scored,
    Away_Late_Goal = LateFH_Goal_Conceded,
    Home_Goals_HT = Period1, # Assuming Period1 is Home Goals
    Away_Goals_HT = Period2 # Assuming Period2 is Away Goals
  )

model <- multinom(
  Match_Outcome ~
    Home_Late_Goal +
    Away_Late_Goal +
    Home_Goals_HT +
    Away_Goals_HT +
    Standard_League_Name,
  data = df_model
)

```

```

## # weights: 30 (18 variable)
## initial value 103576.067963
## iter 10 value 80762.752209
## iter 20 value 77786.838025
## final value 76999.001892
## converged

```

```
cat("\n### Model AIC ###\n")
```

```

##
## ### Model AIC ###

```

```
print(model$AIC)
```

```
## [1] 154034
```

7. Interpretation of Results We calculate Odds Ratios (OR) and P-values to interpret the results.

OR > 1: Increases the likelihood of the outcome (relative to Draw).

OR < 1: Decreases the likelihood.

P-value < 0.05: Statistically significant.

```

# Calculate Z-statistics and P-values
z_stats <- summary(model)$coefficients / summary(model)$standard.errors
p_values <- (1 - pnorm(abs(z_stats))) * 2

# Calculate Odds Ratios
odds_ratios <- exp(coef(model))

print("--- Odds Ratios (Likelihood vs. Draw) ---")

```

```
## [1] "--- Odds Ratios (Likelihood vs. Draw) ---"
```

```
print(round(odds_ratios, 3))
```

```
##      (Intercept) Home_Late_Goal Away_Late_Goal Home_Goals_HT Away_Goals_HT
## Win           1.219           1.753           0.523           1.969           0.671
## Loss          1.161           0.459           1.487           0.702           1.877
##      Standard_League_NameLa Liga Standard_League_NameLigue 1
## Win                        1.041                        1.083
## Loss                       0.879                        1.008
##      Standard_League_NamePremier League Standard_League_NameSerie A
## Win                        1.008                        0.995
## Loss                       0.903                        0.969
```

```
print("--- P-Values ---")
```

```
## [1] "--- P-Values ---"
```

```
print(round(p_values, 4))
```

```
##      (Intercept) Home_Late_Goal Away_Late_Goal Home_Goals_HT Away_Goals_HT
## Win           0e+00           0           0           0           0
## Loss          1e-04           0           0           0           0
##      Standard_League_NameLa Liga Standard_League_NameLigue 1
## Win                        0.3117                        0.0771
## Loss                       0.0017                        0.8582
##      Standard_League_NamePremier League Standard_League_NameSerie A
## Win                        0.8429                        0.9027
## Loss                       0.0123                        0.4779
```